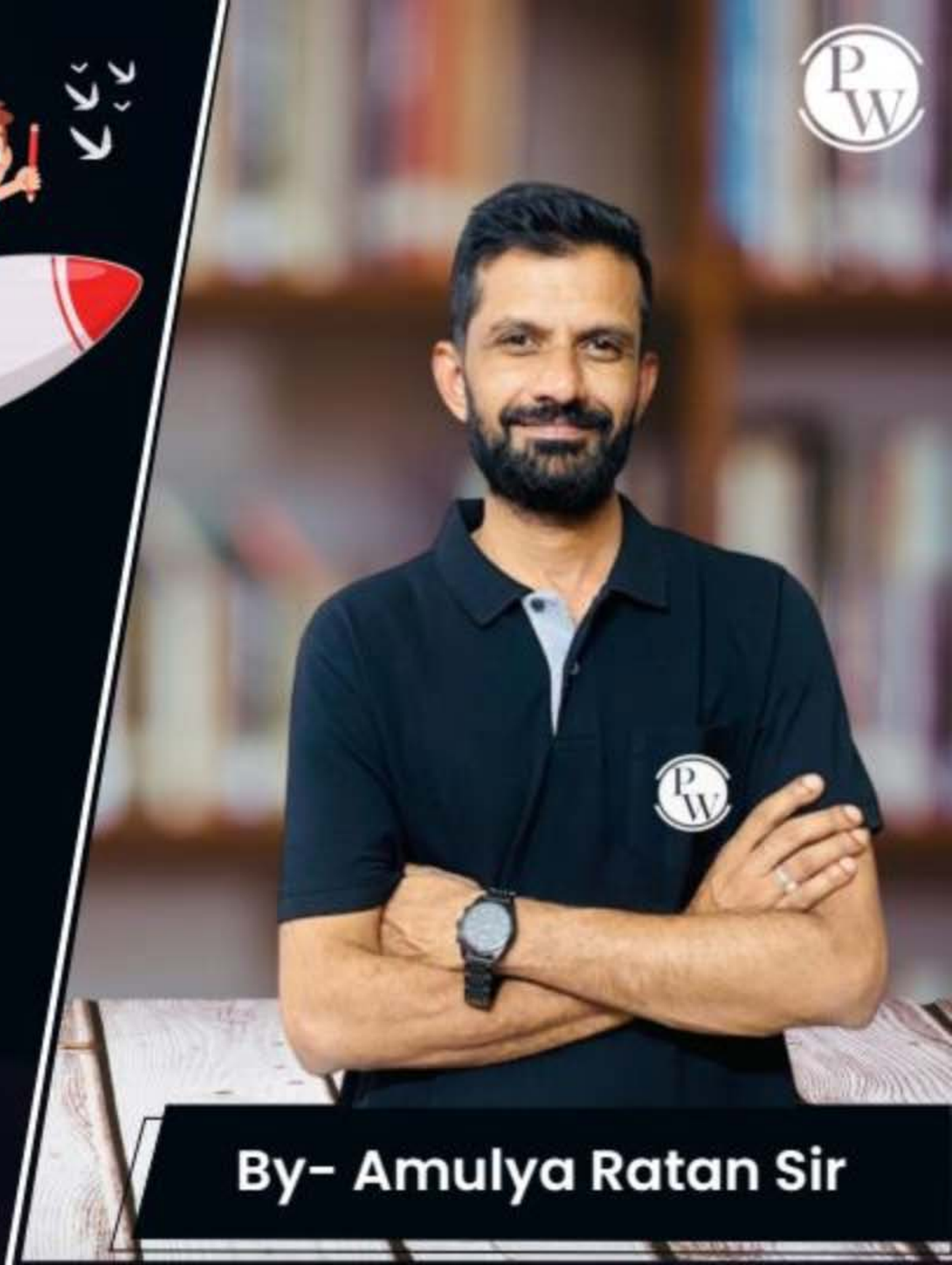
GATE

ALL BRANCHES

General Aptitude

QUANTITATIVE APTITUDE



Lecture No.- 10

By- Amulya Ratan Sir

Recap of Previous Lecture

Topic Time & Distance

Topics to be Covered

Topic-1 Problem on Trains

Topic-2 Boats and Streams

Topic-3 Races

[MCQ]

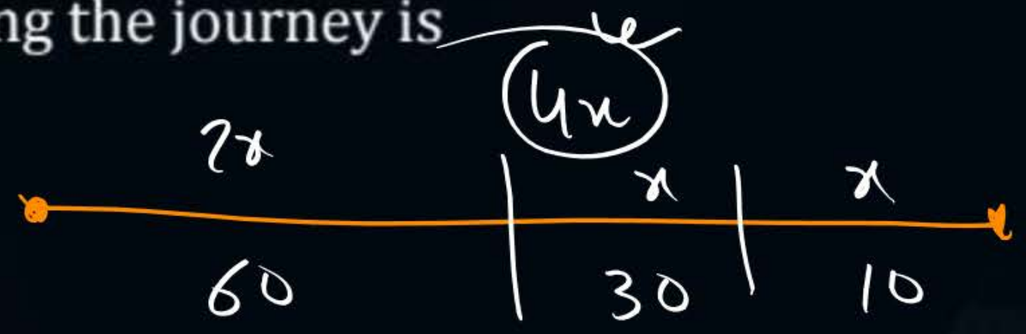
#Q. A tourist covers half of his journey by train at 60 km/hr, half of the remainder by bus at 30 km/hr and the rest by cycle at 10 km/hr. Average speed of the tourist during the journey is

A. 36 km/hr

B. 33 km/hr

C. 24 km/hr

D. 18 km/hr



$$\frac{2x}{60} + \frac{x}{30} + \frac{x}{10} = \frac{4x}{A.S}$$

$$\frac{10x}{60} = \frac{4x}{A.S}$$

Assignment

$A.S. = 6 \times 4$

$= 24$ km/hr

[MCQ]

#Q. A car driver driving at a speed of 68 km/hr locates a lorry 40 meters ahead of him. After 10 seconds, the lorry is 60 meters behind him. Find the speed of the lorry.

$S.C - S.L = 36$

$68 - 36 = S.L$

$D = 100\,m$

$T = 10$ second

$S = 10\,m/sec \times \frac{18}{5}$

$S.L = 32\,km/hr$

$= 36\,km/hr$

[MCQ]

$D = 817$ lines

#Q. Two boys begin together to write out a booklet containing 817 lines. The first boy starts with first line, writing at the rate of 200 lines an hour and the second boy starts with last line. He writes line 817 and so on backwards proceeding at the rate of 150 lines an hour. At what line will they meet?

$R.S. = 350$ lines/hr

A. 469th

B. 467th

C. 465th

D. 463rd

$\text{Time} = \frac{817 \text{ lines}}{350 \text{ lines}} = \frac{817}{350}$ hr

$200 \times \frac{817}{350} = \frac{163400}{350}$

$= 466.85$

467th

Meeting at a point:

$D_1 + D_2 = T \cdot D$

$D_1 = D_2$

$D = 600$ km

R.S. $= 20$ km/hr

$T = \frac{600}{20}$

$= 30$ hrs

40 kph

60 kph

$D = 600$ km

R.S. $= 100$ km/hr

$T = \frac{600}{100}$

$= 6$ hrs

Meeting at a point:

Two objects moving in opposite directions towards each other:
Distance covered by 1st object + Distance covered by 2nd object = Total distance

Two objects moving in same directions :
Distance covered by 1st object = Distance covered by 2nd object

[MCQ] Relative Speed

$D_1 + D_2 = T.D$

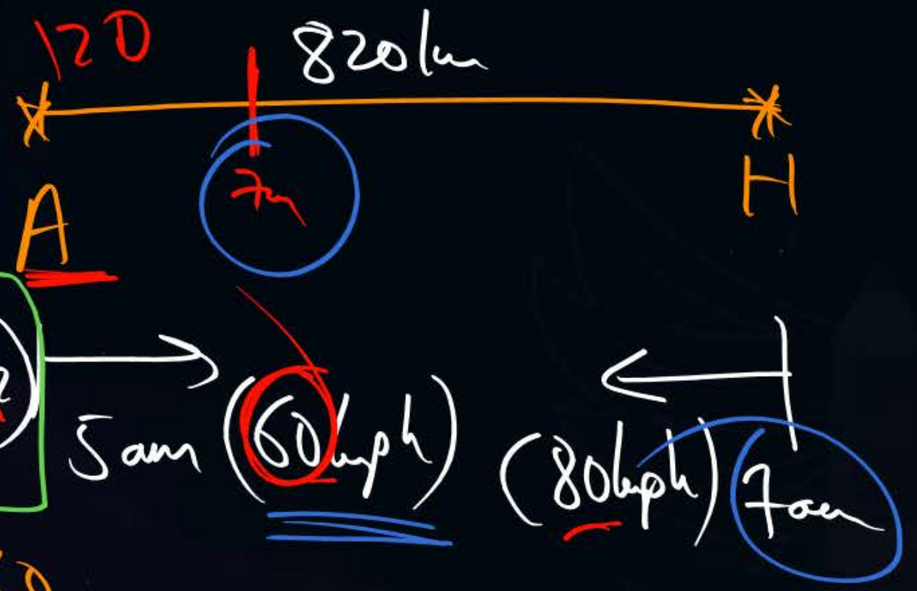
#Q. A train starts from Ahmedabad to Howrah at 5am everyday with the speed of 60kmph. Another train starts from Howrah to Ahmedabad at 7am everyday with the speed of 80 kmph. If the distance between Ahmedabad and Howrah is given as 820km, then at what time the two trains meet each other?

D = 700km

RS = 140kmph

$T = \frac{700}{140} = 5$ hrs

After 5am → 'x' hrs



$820 = 60x + 80(x-2)$

$820 = 60x + 80x - 160$

$7\frac{980}{140} = x$

$x = 7$ hrs

5am + 7hrs

= 12PM

7am + 5hrs

[MCQ]

Relative Speed

$D_1 = D_2$

#Q. A train starts from Chennai to Delhi at 11am with the speed of 110kmph. Another train starts from Chennai to Delhi at 9am the same day with the speed of 90 kmph. If the distance between Chennai and Delhi is given as 1920km, then at what time the two trains meet each other?

D = 180 km

RS = 20 kph

T = 180/20 = 9 hrs

After 9am → 'x' hrs

$90x = 110(x-2)$

$90x = 110x - 220$

$20x = 220$

11 hrs

180 km, 1920 km

C, D

11am (110kph)

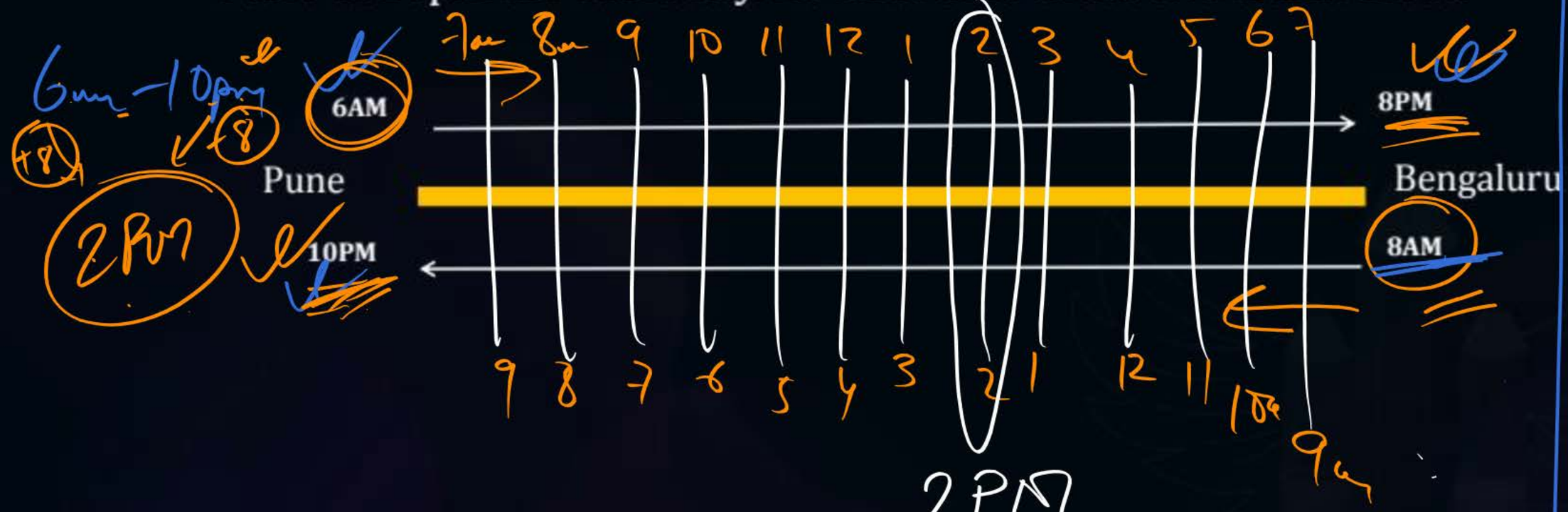
9am (90kph)

9am + 11hrs = 8PM

11am + 9hrs

[MCQ]

#Q. A train starts from Pune at 6am everyday and reaches Bengaluru 8pm the same day. Another train starts from Bengaluru at 8am everyday and reaches Pune at 10pm the same day. At what time these two trains meet?



6am – 10pm

+8

8

2PM

8am – 8pm

+6

-6

2PM

2PM

[MCQ]

8:36 am

$D_1 + D_2 = T.D.$

#Q. A train starts from Leh at 5am everyday and reaches Kanyakumari at 9am the same day. Another train starts from Kanyakumari at 8am everyday and reaches Leh at 2pm the same day. At what time these two trains meet?

$S_{T_1} = \frac{x}{4}$

$S_{T_2} = \frac{x}{6}$

T_1 5am → 'x' km

9am

L K

2pm

After 5am → 'y' hrs

$x = \frac{x}{4}y + \frac{x}{6}(y-3)$

$\frac{y}{4} + \frac{y-3}{6} = 1$

8am

T_2

$\Rightarrow 3y + 2y - 6 = 12$

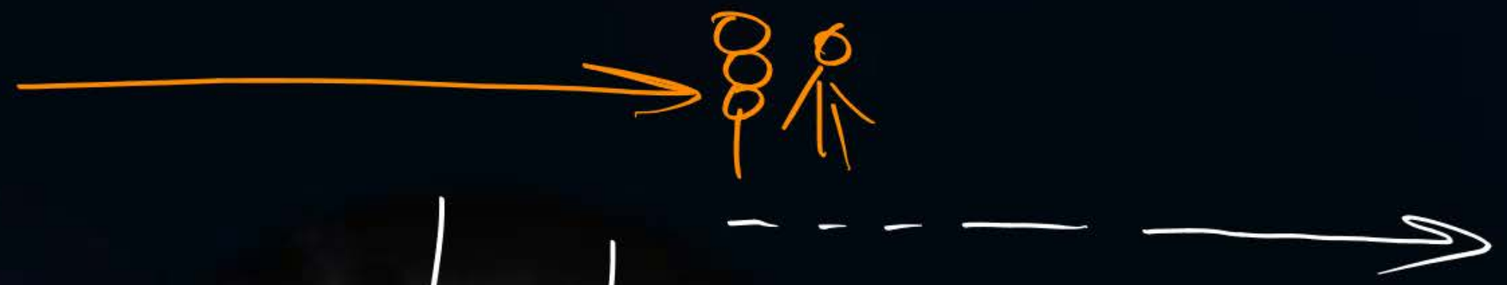
$5y = 18$

$y = \frac{18}{5} = 3\frac{3}{5}$ hrs

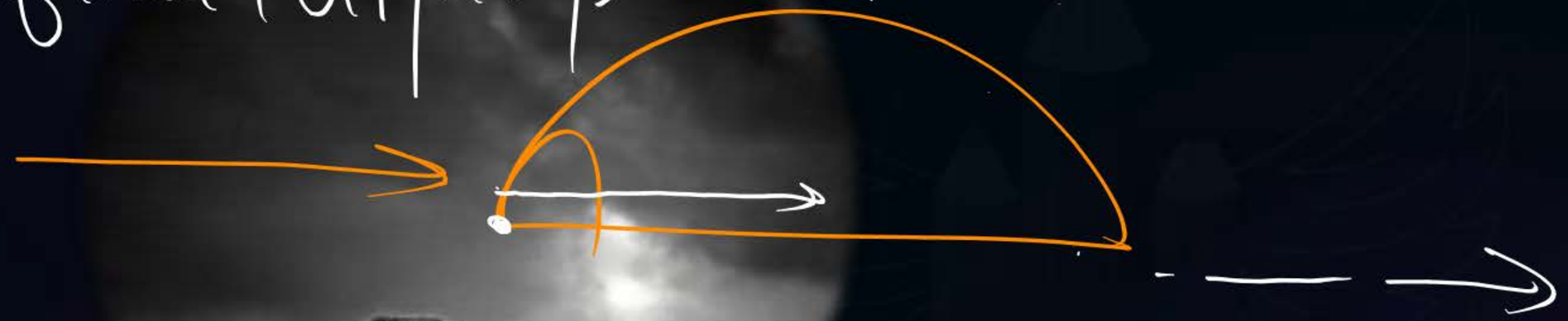
$\frac{3}{5} \times 60$

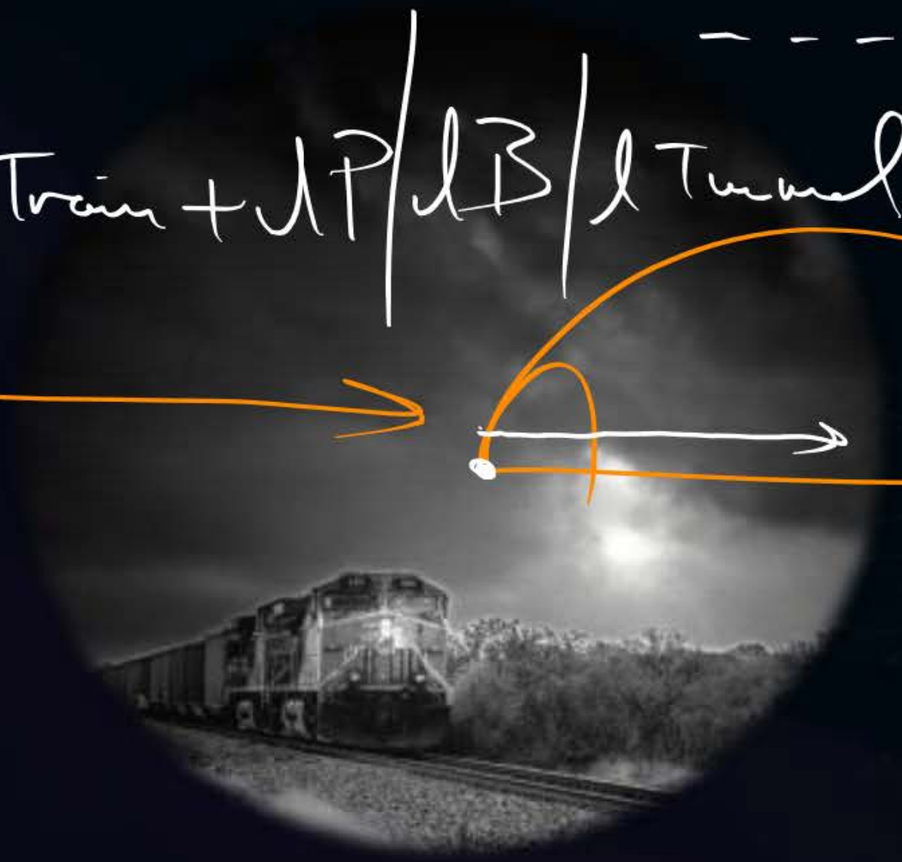
Trains

D = length of Train



D = l of Train + l P / l B / l Tunnel





Trains

When a train try to cross a signal pole or standing man completely...
D = length of the train

R.S = (−)

$$D = l_{T_1} + l_{T_2}$$

When a train try to cross a Platform or Bridge or Tunnel.....
D = length of the train + length of Platform or Bridge or Tunnel

R.S = (+)

Trains

When a train try to cross another moving train.....

SAME DIRECTION	:	Speed (-)
OPPOSITE DIRECTION	:	Speed (+)

Distance = Length of Train 1 + Length of Train 2

[MCQ]

#Q. If a train travelling at 40 km/hr crosses another train of length 100 meter travelling at 14 km/hr in opposite directions in 30 seconds, then find the length of the train.

$D = S \times T$

$l_{T_1} + 100 = 15 \times 30$

$l_{T_1} = 450 - 100$

$l_{T_1} = 350\,m$

$D = l_{T_1} + l_{T_2} = l_{T_1} + 100$

$R.S. = 54\,km/hr \times \frac{5}{18} = 15\,m/sec$

$T = 30$ seconds

[MCQ]

#Q. A train running at 52 km/hr takes 36 seconds to pass a platform. Next it takes 24 seconds to cross a man walking at the platform with 10 km/hr in the same direction. Find the length of the platform.

Train Crossing Platform

$D = l_T + l_P$

$= 280 + l_P$

$S = 52$ kph

$T = 36$ sec

$280 + l_P = 52 \times \frac{5}{18} \times 36$

$l_P = 520 - 280 = 240$ m

Train crossing Man

$D = l_T$

R.S. = 42 km/hr

$T = 24$ sec

$l_T = 42 \times \frac{5}{18} \times 24$

$l_T = 280$ m

Boats & Streams

$(S \cdot B) - (S \cdot W) =$ Upstream Speed (US)

$(S \cdot B) + (S \cdot W) =$ Downstream Speed (DS)

8 km/hr

$$S \cdot B = \frac{D \cdot S + U \cdot S}{2}$$

$$S \cdot W = \frac{D \cdot S - U \cdot S}{2}$$

10 km/hr

6 km/hr

2 km/hr

[MCQ]

#Q. A man rows his boat downstream @ 18 km/hr & upstream @ 10 km/hr. Find the speed of boat in still water.

$$S \cdot B = \frac{D \cdot S + U \cdot S}{2} = \frac{18+10}{2} = 14 \text{ km/hr}$$

A. 12 km/hr

B. 10 km/hr

C. 14 km/hr

D. 16 km/hr

[MCQ]

#Q. A boat traveled from A to B and back to A from B in 5 hours. If the speed of boat in still water and the speed of stream be 7.5 kmph and 1.5 kmph, then what is the distance between A and B?

'x' km

$T = \frac{D}{S}$

A 18 km

B 10 km

C 16 km

D 5 km

$$\frac{x}{9} + \frac{x}{6} = 5 \Rightarrow \frac{5x}{18} = 5$$

$\therefore x = 18$ km

Races

$$\frac{A}{B} \times \frac{B}{C} = \frac{1000}{900} \times \frac{1000}{950}$$

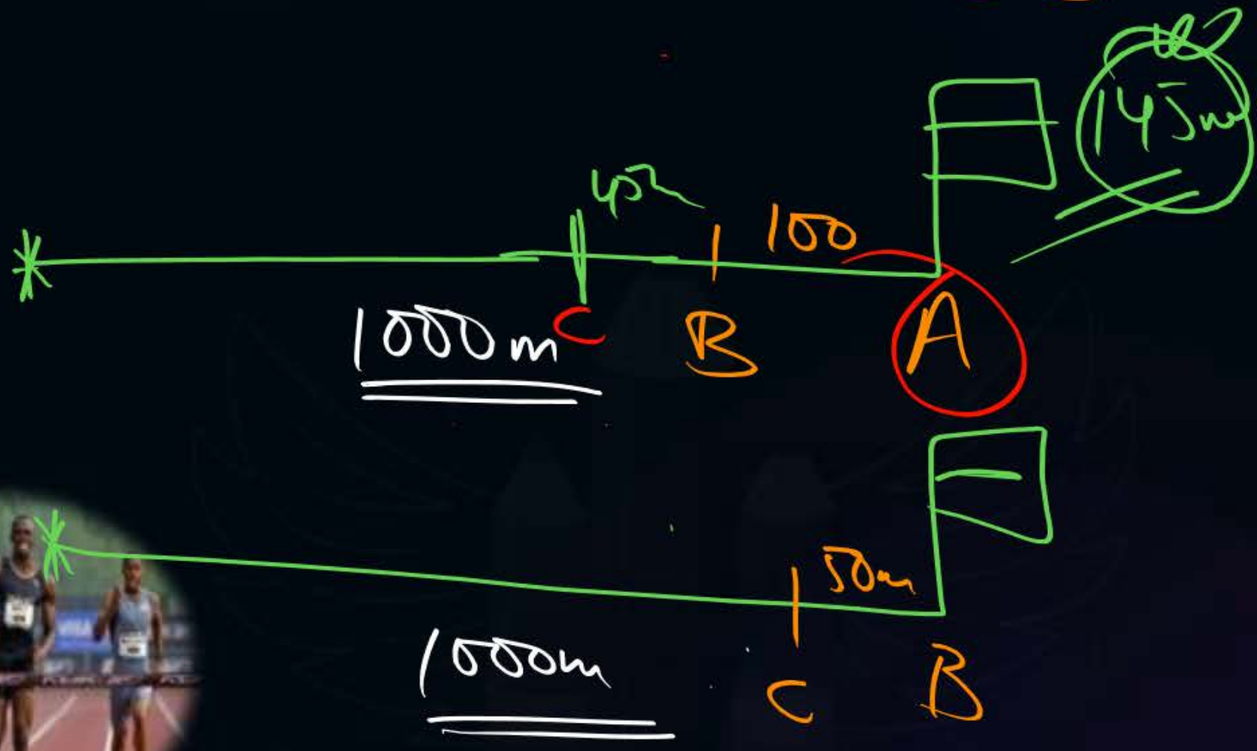
$$\frac{A}{C} = \frac{1000}{855}$$

$$\frac{1000}{855}$$

#Q. In a km race A defeats B by 100 meters and B defeats C by 50 meters, then A defeats C by how many meters?

$$\frac{A}{B} = \frac{1000}{900}$$

$$\frac{B}{C} = \frac{1000}{950}$$



[MCQ]

#Q. In a km race A defeats (beats) B by 200 meters and B defeats C by 100 meters, then A defeats C by how many meters?

$$\frac{A}{C} = \frac{1000}{800} \times \frac{1000}{900}$$

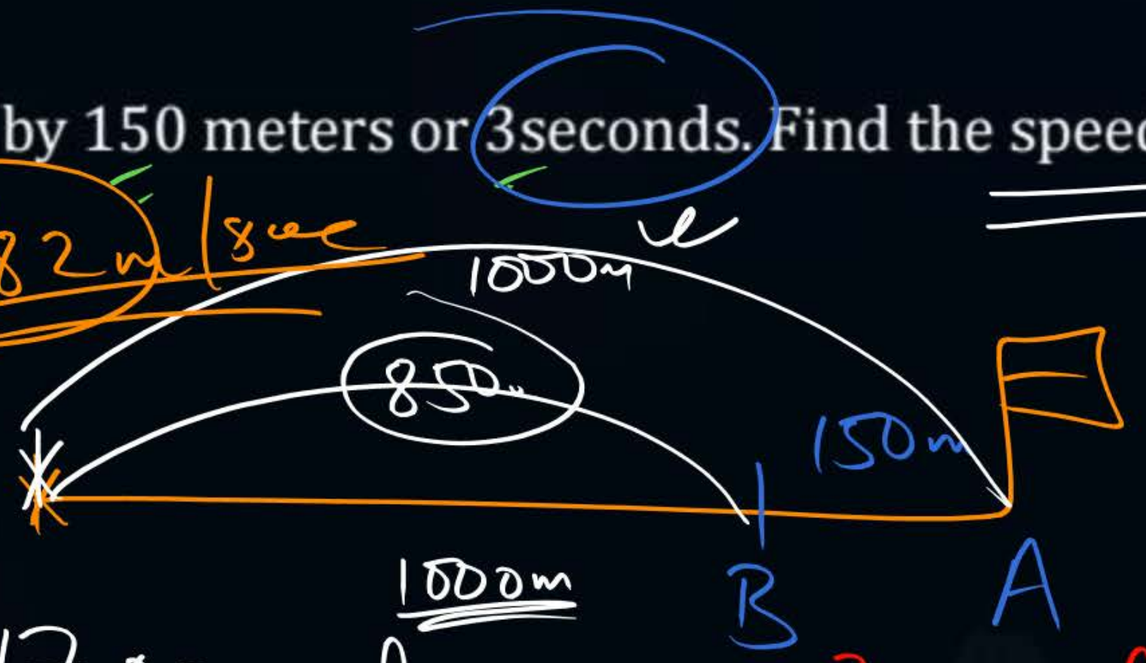
$$\frac{A}{C} = \frac{1000}{720}$$

280m

[MCQ]

#Q. In a km race A defeats B by 150 meters or 3seconds. Find the speed of A.

$S_A = \frac{1000}{17} = 58.82\ m/sec$



$\frac{850}{50} = 17$ second

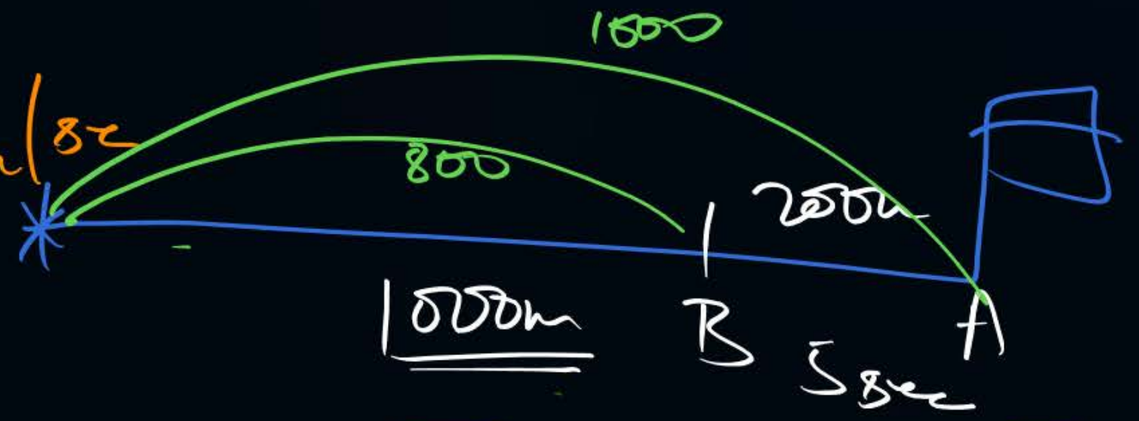
3 seconds

$S_B = \frac{150}{3} = 50\ m/sec$

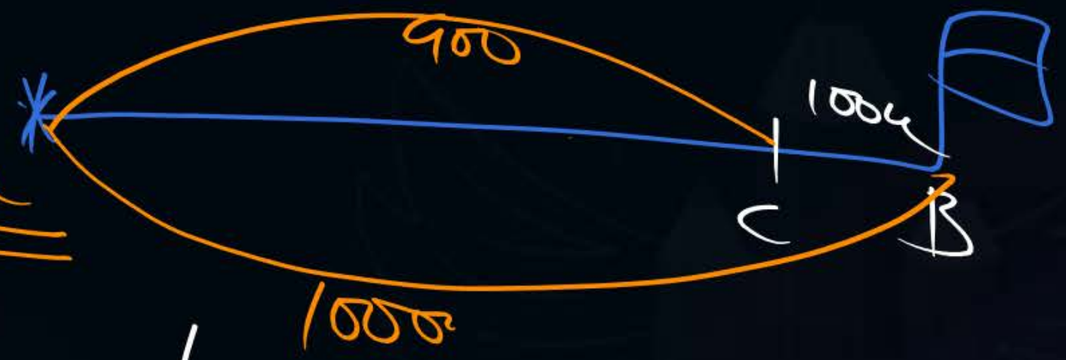
[MCQ]

#Q. In a km race A defeats B by 200 meters or 5seconds whereas B defeats C by 100m. Find the speed of C.

$S_B = \frac{200}{5} = 40\,m/sec$



$\frac{1000}{40} = 25\,sec$



$S_C = \frac{900}{25} = 36\,km/sec$

2 mins Summary

Topic Trains, Boats & Races

THANK - YOU